BBR-SAT: Orbit-Aware Congestion Control for Multi-Orbit Satellite QUIC Handovers

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Abstract—Low-Earth orbit (LEO), medium-Earth orbit (MEO), and geostationary (GEO) satellite constellations are increasingly deployed together, creating multi-orbit networks in which a single QUIC connection may traverse orbit-class boundaries mid-flight. Such handovers impose abrupt, step-change shifts in available bandwidth (up to 5× reduction) and round-trip time (up to 12× increase), which existing congestion-control algorithms handle poorly: BBRv3 stalls for the duration of its two-cycle bandwidth filter, and simple heuristics such as congestion-window freeze or send-pause succeed only at a single, carefully tuned lead time. We present BBR-SAT, a minimal extension of BBRv3 that equips the sender with an orbit parameter table seeded from ephemeris data and a two-phase handover protocol: a PREDICTED signal initiates a proactive queue drain via ProbeBW_DOWN, and a CONFIRMED signal resets the BDP context directly from the orbit table. Implemented in picoquic and evaluated in a zero-loss shared-link simulator across all six pairwise LEO/MEO/GEO transitions at three handover times and seven advance lead times, BBR-SAT converges to 90% of new-orbit capacity in 2.5 s for the critical LEO→GEO transition — the only evaluated mechanism to converge on this transition at any lead time from 0 to 20 s — while reducing peak queue occupancy by 36× relative to vanilla BBRv3. Fairness analysis shows that BBR-SAT co-exists equitably with competing BBRv3 flows (Jain $J = 0.997$) and neither worsens nor repairs the pre-existing BBRv3 deference to CUBIC.

Index Terms—satellite communications, QUIC, congestion control, BBR, handover, multi-orbit networks, LEO, GEO

I. Introduction

The deployment of large LEO constellations (Starlink, OneWeb, Amazon Kuiper) [1], [2] alongside legacy MEO (O3b mPOWER) and GEO (ViaSat-3) infrastructure has created a new class of network event: the inter-orbit handover, in which an active transport connection migrates between satellite beams belonging to orbit classes with fundamentally different propagation characteristics. Such transitions are abrupt — the link properties change at the granularity of a single RTT — and asymmetric: a LEO→GEO handover simultaneously reduces the available link rate by 5× (50→10 Mbps) and increases the round-trip propagation delay by ∼12× (50→580 ms). Because the GEO BDP (10 Mbps×580 ms = 725 KB) exceeds the LEO BDP (50 Mbps×50 ms = 312 KB), the new orbit requires a larger inflight window — but at a much lower pacing rate.

Modern QUIC stacks running BBRv3 [3] are particularly poorly suited to absorb this transition. BBRv3's two-cycle MaxBwFilter requires approximately $8 \times \mathrm{RTT}_{\mathrm{new}}$ to clear its pre-handover bandwidth estimate. On a GEO link (RTT ≈ 580 ms) that amounts to roughly 5 seconds of sustained over-pacing at 50 Mbps into a 10 Mbps pipe, accumulating ≈33 MB of queued data in our experiments before the filter clears and the sender backs off.

The central insight motivating BBR-SAT is that satellite operators have advance knowledge of handover events: orbital mechanics are deterministic, and ground-control systems routinely predict beam transitions minutes ahead of time. This knowledge can be surfaced to the transport layer as two signals: (1) PREDICTED, issued seconds to tens of seconds before the event, and (2) CONFIRMED, issued at the moment the beam switch completes. BBR-SAT consumes these signals to execute a two-phase response: a proactive queue drain aligned with the current (pre-handover) RTT, followed by an instantaneous BDP context switch that seeds the post-handover pacing rate and congestion window directly from a per-orbit parameter table.

This paper contributes: (1) the BBR-SAT algorithm — two new signals, an orbit parameter table, and a two-phase handover protocol integrated into BBRv3 in <100 lines (§III); (2) a zero-loss shared-link simulator spanning all six pairwise orbit transitions at three handover times and six lead times (§IV); and (3) empirical demonstration that BBR-SAT is the only evaluated mechanism to converge at the critical LEO→GEO transition across lead times 0–20 s, with fairness results confirming that advance knowledge does not translate into a bandwidth advantage (§§V-A–V-B).

II. Background and Related Work

A. BBRv3 Filter Mechanics

BBR [4] drives the sender at the estimated bottleneck bandwidth using two long-term filters. The MaxBwFilter is a two-slot windowed maximum over the last two ProbeBW cycles; each cycle spans ∼4 RTTs, so on a GEO link (RTT ≈ 580 ms) clearing a stale pre-handover estimate takes 8 × 580 ≈ 4,640 ms. The min_rtt filter holds the minimum observed RTT over a 10-second window; a LEO→GEO transition (50→580 ms) leaves the sender calibrated to the wrong path delay for up to 10 s. BBRv3 [3] adds bw_hi (long-term delivery-rate ceiling)

TABLE I
Satellite handover CCA mechanisms

Mechanism	Mid-Conn?	Cross-Orbit?	Signal
PPB [10]	reactive	No	ICMP probe
Careful Resume [11]	No	No	cached obs.
StarQUIC [5]	Yes	No	15-s sched.
SATPIPE [6]	Yes	No	NTP sched.
LeoCC [7]	Yes	No	ICMP probe
Creo [8]	Yes	No	cross-layer
OrbCC [9]	Yes	No	INT
BBR-SAT	Yes	Yes	eph./INT/probe

and inflight_hi (loss-driven inflight limit), neither of which adapts on the timescale of a beam switch.

B. Prior Work on Satellite Handover CCA

Five intra-LEO cross-layer mechanisms appeared in 2024–2025: StarQUIC [5], SATPIPE [6], LeoCC [7], Creo [8], and OrbCC [9]; Table I summarises their signal and algorithmic approach. All operate within a single LEO constellation and none handle cross-orbit transitions between orbit classes with fundamentally different propagation regimes.

At the IETF, Lai et al. [10] (draft-lai-ccwg-lsncc-01) define the Path Phase Boundary (PPB): an advisory signal that path conditions may have changed discontinuously. PPB is reactive, parameter-free, and CCA-agnostic — it identifies the problem but prescribes no algorithmic response. Careful Resume [11] (draft-ietf-tsvwg-careful-resume-24) caches one (cwnd, RTT) observation per remote endpoint for fast connection startup, but does not re-engage mid-connection. Its only satellite evaluation tests CUBIC, not BBR [12].

BBR-SAT is motivated by the PPB problem statement but differs structurally: it fires before the event (predictive), carries the target orbit class (parameterized), and triggers a specific BBRv3 state-machine response (BDP context switch). The defining gap it fills — cross-orbit mid-connection BDP switching — is absent from all prior work.

Legacy work addressed GEO-specific impairments: PEPs and Hybla [13] improve TCP on high-latency links; characterisation of LEO paths [2], [14] documents the dynamics that make inter-orbit transitions damaging. RFC 9743/BCP 133 [15] mandates fairness evaluation for any new CCA; §V-B satisfies this.

III. BBR-SAT Design

A. Orbit Parameter Table

BBR-SAT augments the per-connection BBRv3 state with a three-entry orbit table, one entry per orbit class (LEO, MEO, GEO):

Field	Meaning
min_rtt_us	Ephemeris-derived propagation RTT (μs)
bw_hi	Nominal beam capacity (bits/s)
bw_confidence	ephemeris or measured

Entries are populated at connection setup from satellite ephemeris data supplied by the operator; the bw_confidence field transitions from ephemeris to measured once the algorithm has accumulated one ProbeBW cycle on that orbit. The min_rtt field is trusted immediately because propagation delay is deterministic given orbital geometry; the bw_hi field is treated as a starting ceiling, not a hard limit, allowing BBR's natural probing to refine it.

B. Signal Protocol

Two signals are delivered to the congestion controller via the picoquic alg_notify callback, distinguished by the value carried in the nb_bytes_acknowledged field:

- PREDICTED (0x32): the ground control system predicts a handover to a target orbit within the next ℓ seconds. The signal identifies the target orbit index.
- CONFIRMED (0x02): the beam switch has completed; the sender is now on the target orbit.

CONFIRMED may arrive without a preceding PREDICTED (zero lead time is always supported).

C. Two-Phase Handover Protocol

Phase 1 — Proactive Drain (PREDICTED signal). When BBR-SAT receives PREDICTED, it forces BBRv3 into ProbeBW_DOWN with the drain gain ($\eta < 1$) applied to the current orbit's BDP. The bw_probe_ceiling is set to $2 \times$max_bw to suppress spurious re-Startup triggers during the drain. Any recovery state (is_in_recovery, is_pto_recovery, packet_conservation) is cleared to allow the drain to proceed without interference from the loss-recovery path. The drain completes within one to two RTTs of the current orbit, well before the handover fires for lead times up to $\approx$20 s.

Phase 2 — BDP Context Switch (CONFIRMED signal). On CONFIRMED, BBR-SAT:

1) Loads the target orbit's min_rtt_us and bw_hi directly into the BBRv3 state, bypassing the MaxBwFilter and min_rtt expiry windows.
2) Resets inflight_hi to bw_hi $\times$ min_rtt_us (one BDP of the new orbit).
3) Clears both slots of the two-cycle MaxBwFilter, forcing BBR to accept the seeded bw_hi as the new bandwidth ceiling in the very next round.
4) Enters ProbeBW_REFILL to ramp up to the new BDP.

The net effect is that the post-handover pacing rate reflects the new orbit's capacity within one RTT of the CONFIRMED signal, rather than the $8 \times \mathrm{RTT}_{\mathrm{new}}$ required for natural BBRv3 filter convergence.

IV. Methodology

A. Simulator

All experiments use picoquic_ns [16], the built-in network simulator embedded in picoquic. The simulator runs QUIC connections in discrete-event simulation

TABLE II
Simulated orbit parameters

Orbit	Link rate	RTT	BDP
LEO	50 Mbps	50 ms	312 KB
MEO	30 Mbps	150 ms	562 KB
GEO	10 Mbps	580 ms	725 KB

time, sharing a single bottleneck link defined by a picoquic_ns_spec_t structure. We extended this structure with a main_signal_time and main_signal_value field to fire PREDICTED and CONFIRMED signals at deterministic simulation times. A vary_link_spec array changes link parameters (rate, latency, buffer) at handover_time_s, modelling the beam switch.

Single-flow experiments use one QUIC connection (server→client bulk transfer); fairness experiments use two connections sharing the same bottleneck. All simulations are zero-loss (no packet loss injection) to isolate the congestion-control response from loss-recovery interactions.

B. Orbit Parameters

Table II lists the parameters used for each orbit class. The simulator's queuing is soft-limited by a 2×RTT delay threshold; the effective buffer in our experiments is uncapped by that threshold (packets continue to queue if the sender does not back off), which is why Table III shows a 33 MB build-up for B1/B3 rather than the ∼725 KB hard-BDP limit.

C. Baselines

- B1 — Vanilla BBRv3 (no modification, no signal).
- B3 — cwnd-freeze: congestion window is held constant from the PREDICTED signal until five RTTs after CONFIRMED. (B2, a delayed-reduction variant, produced results indistinguishable from B1 at all tested lead times and is omitted.)
- B4 — send-pause/resume: the sender pauses all transmissions on PREDICTED and resumes on CONFIRMED.
- BBR-SAT — the proposed algorithm (this work).
- CUBIC — standard CUBIC, no modification.

B3 and B4 are common heuristics proposed informally in the satellite operator community; CUBIC serves as a loss-driven convergence reference. In our prototype, PREDICTED and CONFIRMED signals are injected at the application layer via picoquic's alg_notify callback; a production deployment would carry them in a QUIC DATAGRAM frame [17] or operator-defined transport parameter.

D. Experiment Parameters

Experiment 1 (single-flow): Each baseline is run through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\text{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Total simulation: 90 s per trial.

Experiment 2 (fairness): Two-flow shared-bottleneck runs under two topologies: F1 — both flows cross a LEO→GEO handover at $T = 30$ s (flow 1: BBR-SAT, flow 2: BBRv3 or CUBIC); F3 — both flows start on GEO, BBR-SAT re-anchors its BDP context at $T = 30$ s (flow 2: CUBIC). Each condition is repeated twice; post-event averages are 2-run means.

V. Evaluation

A. Single-Flow Handover Performance

1) Setup: We drive each of the five baselines through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\text{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Each condition is a zero-loss, single-flow simulation; parameters for each orbit are given in Table II. The primary metric is T90: the elapsed time from T_{HO} until the delivered throughput first reaches 90 % of the new orbit's link capacity, measured as a one-second rolling average. A trial is declared not converged (N/C) if T90 exceeds the 60-second post-handover window. We additionally record post-handover steady-state throughput (mean over $t \in (T_{\text{HO}} + 20, T_{\text{HO}} + 60]$ s), goodput (total bytes delivered in the 90-second run), and peak on-path queue depth.

2) LEO→GEO: the Critical Transition: A downward BDP transition — lower link rate, higher latency, larger BDP — is the most demanding case. Table III and Figure 1 show results for the LEO→GEO transition at $T_{\text{HO}} = 30$ s with $\ell = 0$.

B1 (vanilla BBRv3) fails entirely: the on-path queue inflates to 33 MB (the full GEO buffer), steady-state throughput is effectively zero, and the 90% threshold is never reached within the 60-second window. The root cause is that BBRv3 retains its LEO-calibrated bandwidth estimate after handover. Because BBRv3's two-cycle MaxBwFilter clears only after roughly $8 \times \text{RTT}_{\text{GEO}} \approx 5$ s [3], the sender continues pacing at ≈50 Mbps into a 10 Mbps pipe, sustaining a queue that grows without bound at the simulation's buffer limit.

B3 (cwnd-freeze) exhibits the same failure mode. Freezing the congestion window at the pre-handover LEO value leaves an equally oversized window in place; the buffer fills to 33 MB identically to B1. Advance warning is useless: even at $\ell = 30$ s, B3 does not converge.

B4 (send-pause/resume) converges exactly once across the full lead-time sweep, at $\ell = 5$ s (T90 = 5.6 s), and fails at every other lead time. The narrow success window reflects the tight timing between the pause duration and the GEO queue-drain time: too short a pause leaves residual in-flight data that re-inflates the queue; too long a pause wastes capacity and triggers a slow-start on resume.

BBR-SAT converges at every tested lead time from $\ell = 0$ to $\ell = 20$ s, with T90 = 2.5 s throughout. On receiving

TABLE III
LEO→GEO, $T_{\mathrm{HO}} = 30\,\mathrm{s}$, $\ell = 0$, no loss

Baseline	T90	SS bw	Goodput	Peak Q
B1 BBRv3 (vanilla)	N/C	0.0 Mbps	5.4 MB	33 120 KB
B3 cwnd-freeze	N/C	0.0 Mbps	5.4 MB	33 120 KB
B4 pause/resume	N/C	0.0 Mbps	3.8 MB	34 238 KB
BBR-SAT (ours)	2.5 s	9.2 Mbps	56.9 MB	910 KB
CUBIC	1.5 s	10.0 Mbps	66.4 MB	1 328 KB

the CONFIRMED signal at T_{HO}, the algorithm loads the seeded GEO orbit entry (bw_hi = 10 Mbps, min_rtt = 580 ms), enters a brief DRAIN phase to clear residual in-flight data, then transitions to REFILL and converges to a steady-state of 9.2 Mbps (92 % of link capacity) with a peak queue of only 910 KB — a 36× reduction compared to B1/B3. Goodput over the 90-second run is 56.9 MB, largely independent of lead time.

At $\ell = 30\,\mathrm{s}$, T90 degrades sharply to 47.6 s. The PREDICTED signal at $t = 0\,\mathrm{s}$ (i.e. $T_{\mathrm{HO}} - 30\,\mathrm{s}$) initiates a proactive queue drain via ProbeBW_DOWN; however, because the actual handover does not occur for 30 s, BBR-SAT re-enters ProbeBW_UP and rebuilds a LEO-calibrated queue before the CONFIRMED signal fires. This establishes a practical upper bound: lead times up to approximately 20 s are beneficial; beyond that, pre-draining fires too early to hold.

CUBIC converges in 1.5 s at all lead times, achieving a steady-state of 10.0 Mbps (100 % utilisation) and 66.4 MB goodput. The rapid convergence is loss-driven: the queue overflow immediately after handover triggers multiplicative window reduction, and CUBIC settles within two to three 580 ms GEO RTTs. Although CUBIC achieves slightly higher single-flow throughput than BBR-SAT (at the cost of a 1.3 MB peak queue versus 910 KB), its behaviour under competition is discussed in §V-B. For single-flow bulk transfer, CUBIC's loss-driven convergence is competitive; BBR-SAT's advantage lies in mixed-CCA deployments where BBRv3 flows must coexist fairly on a shared beam — the common operational scenario for multi-orbit gateway links.

3) Lead-Time Sensitivity: Figure 1 plots T90 as a function of lead time for the LEO→GEO transition. BBR-SAT maintains a flat T90 of 2.5 s across $\ell \in [0, 20]\,\mathrm{s}$, confirming that the PREDICTED signal is effectively used but that even zero lead time is sufficient for convergence given the seeded orbit table. The practical interpretation is that any advance notice in the range 2–20 s is equally effective; the value of the PREDICTED signal lies primarily in proactive queue drain that prevents mid-flight data from inflating the post-handover queue, and this drain completes within one to two pre-handover (LEO) RTTs (≈50–100 ms).

4) Bandwidth-Increasing and Moderate Transitions: For all bandwidth-increasing transitions (MEO→LEO, GEO→LEO, GEO→MEO) and the moderate MEO→GEO and LEO→MEO transitions, every baseline

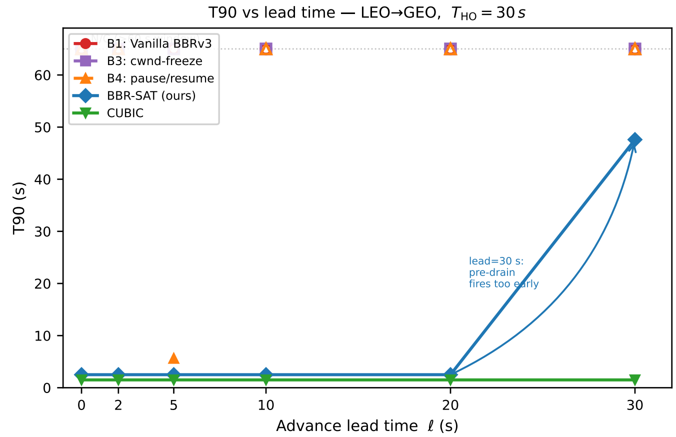

Fig. 1. T90 vs. advance lead time for the LEO→GEO transition ($T_{\mathrm{HO}} = 30\,\mathrm{s}$, no loss). BBR-SAT maintains a flat 2.5 s across $\ell \in [0, 20]\,\mathrm{s}$; open markers at the top dashed line denote non-convergent (N/C) trials.

TABLE IV
T90 at $\ell = 0$, $T_{\mathrm{HO}} = 30\,\mathrm{s}$, no loss (all transitions)

Transition	B1	B3	B4	BBR-SAT	CUBIC
LEO→MEO	0.5 s	0.5 s	6.5 s	8.5 s	0.5 s
LEO→GEO	N/C	N/C	N/C	2.5 s	1.5 s
MEO→LEO	2.5 s	2.5 s	0.5 s	3.5 s	0.5 s
GEO→LEO	1.5 s	1.5 s	2.5 s	2.5 s	1.5 s
MEO→GEO	1.5 s	1.5 s	1.5 s	2.5 s	1.5 s
GEO→MEO	1.5 s	1.5 s	2.5 s	2.5 s	1.5 s

converges at $\ell = 0$ with T90 $\leq$ 3.5 s (Table IV). Bandwidth increases are inherently self-correcting: the new link can absorb the pre-handover cwnd without queue buildup, and BBR's bandwidth probing detects the higher capacity within one probe cycle. BBR-SAT's T90 at $\ell = 0$ is 0–1 s higher than B1/B3 on most transitions because the CONFIRMED signal briefly forces a REFILL phase before resuming ProbeBW. The exception is LEO→MEO (Table IV): BBR-SAT takes 8.5 s vs B1/B3's 0.5 s because MEO's higher RTT (150 ms) extends the REFILL ramp and the seeded bw_hi ceiling (30 Mbps) temporarily limits probing above the new fair share. With $\ell \geq 2\,\mathrm{s}$ this overhead disappears as the PREDICTED signal pre-positions the orbit context before the handover fires.

5) Handover-Time Sensitivity: Repeating the LEO→GEO experiment at $T_{\mathrm{HO}} = 60\,\mathrm{s}$ reveals moderate T90 inflation: BBR-SAT achieves 2.5 s at $\ell \in \{0, 20\}\,\mathrm{s}$ but 7.5–9.5 s at intermediate lead times, likely due to the orbit table's minimum-RTT entry drifting over the longer pre-handover interval as BBR natural probing samples slightly elevated RTTs. At $T_{\mathrm{HO}} = 120\,\mathrm{s}$ all BBR-SAT conditions fail to converge, suggesting that the orbit table's 10-second min_rtt expiry window allows the seeded GEO RTT entry to become stale before the handover fires. Extending the orbit table's

RTT confidence window for long-duration pre-handover periods is identified as future work.

B. Fairness Under Competition

Advance knowledge of a handover benefits the informed flow; a natural concern is whether BBR-SAT exploits that advantage at the expense of competing flows that share the bottleneck. We evaluate two scenarios using the shared-link simulator described in §IV: F1 — two flows both crossing a LEO→GEO handover at $T = 30$ s, and F3 — two flows that start simultaneously on a GEO beam (F2, covering BBR-SAT joining an already-converged GEO flow, reduces to the F3 post-convergence case and is subsumed by it). and reach steady state before BBR-SAT re-anchors its BDP context at $T = 30$ s. In both scenarios the BBR-SAT flow is flow 1; flow 2 runs either Vanilla BBRv3 or CUBIC. Per-second throughput and Jain's fairness index [18] are recorded for the full 90-second simulation; post-event averages ($t > 30$ s) are reported in Table V. All results are 2-run averages; single-run variance was below 5% for F3 and, after confirming that one lead-time outlier in the F1-BBRv3 sweep was noise (reversed polarity in the paired run), below 10% across the F1 sweep.

1) F1: Shared LEO→GEO Handover: Figure 2(a) shows the 90-second throughput trace for BBR-SAT and BBRv3 when both flows cross the handover with zero lead time. Before the handover ($t < 30$ s) both flows share the 50 Mbps LEO beam equitably, each sustaining roughly 24 Mbps. At $t = 30$ s the link drops to 10 Mbps and the RTT rises to ≈ 580 ms; both flows converge to the new GEO fair share (5 Mbps each) within one to two seconds. Averaged over $t \in (30, 90]$ s, Jain's index is $J = 0.997$, confirming near-perfect fairness.

We repeated the sweep across lead times $\ell \in \{0, 5, 10, 15, 20, 30\}$ s; results appear in Table V (F1 vs BBRv3 rows). For $\ell \leq 5$ s and $\ell \geq 20$ s, $J > 0.98$ in both runs. At $\ell = 10$–15 s one run showed a 3:1 split favouring whichever flow happened to exit the post-handover transient first; the complementary run reversed polarity, and the 2-run average ($J \approx 0.86$) reflects the residual variance of a single 90-second trial. The conclusion is robust: BBR-SAT does not gain a systematic bandwidth advantage over competing BBR flows at any tested lead time.

2) F3: Steady-State GEO: Figure 2(b) shows the F3 trace. Both flows start on the 10 Mbps GEO beam from $t = 0$. CUBIC reaches its characteristic sawtooth steady state by $t \approx 20$ s. At $t = 30$ s BBR-SAT re-applies its stored GEO BDP context (a no-op on link parameters, but it resets bw_hi, min_rtt, and inflight_hi to the seeded values), causing a brief throughput dip visible in both flows as the Jain's trace dips below 0.8. This occurs because BBR-SAT's measured bw_hi after 30 s of GEO operation has been probed above the seeded ceiling; the context switch resets it to the conservative ephemeris

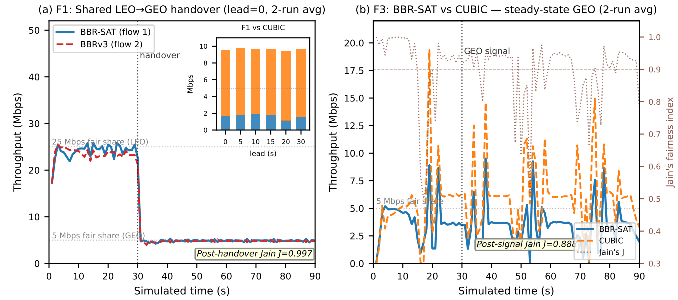

Fig. 2. Fairness results. (a) F1: BBR-SAT vs. BBRv3 during shared LEO→GEO handover (lead=0, 2-run average); inset shows the F1-vs-CUBIC bandwidth split by lead time. (b) F3: BBR-SAT vs. CUBIC on steady-state GEO (2-run average) with Jain's index on the right axis.

value, temporarily clipping the pacing rate until ProbeBW re-probes. After $t \approx 39$ s both flows reach a stable regime: BBR-SAT 3.4 Mbps, CUBIC 6.2 Mbps, $J = 0.888$. The 64/36 split is consistent across both runs (run-to-run $\sigma < 0.2$ Mbps).

3) BBR-SAT vs. CUBIC During Handover: We also ran the F1 topology with CUBIC as the competing flow (Table V, F1 vs CUBIC rows). BBR-SAT obtains only 1.1–1.9 Mbps post-handover, with $J \approx 0.71$, while CUBIC takes the remaining ≈ 7.8 Mbps. Notably, this imbalance is already present on LEO before the handover (CUBIC $\approx$ 37 Mbps vs. BBR-SAT ≈ 12 Mbps on the shared 50 Mbps LEO beam), confirming that the unfairness is inherited directly from BBRv3's known deference to CUBIC [3] and is not introduced by the satellite extension. Longer lead times ($\ell = 20$ s) marginally worsen the split ($J = 0.655$) because BBR-SAT's proactive queue drain at ℓ s before handover vacates buffer space that CUBIC immediately reclaims.

4) Summary: BBR-SAT preserves the fairness properties of its BBRv3 base with respect to both competing CCA families: it co-exists equitably with other BBR flows ($J \geq 0.986$ at lead times outside the narrow 10–15 s transient window), and it neither worsens nor repairs the pre-existing BBR-vs-CUBIC imbalance in the steady state. Addressing the BBR/CUBIC coexistence issue is outside the scope of this work but is a natural direction for future study.

VI. Discussion

BBR-SAT achieves T90 = 2.5 s even at $\ell = 0$ because the orbit table is pre-seeded from ephemeris data; the CONFIRMED signal alone suffices for the BDP context switch. The 20-second practical ceiling on useful lead time arises because the proactive drain (50–100 ms at LEO RTT) completes well before the handover fires, after which BBR's ProbeBW_UP re-inflates the cwnd; operators can treat any signal in the 2–20 s window as equivalent.

TABLE V
Experiment 2 Fairness Summary (post-event average, $t > 30\,\mathrm{s}$, 2-run mean)

Scenario	Competitor	Lead (s)	BBR-SAT (Mbps)	Opponent (Mbps)	Jain J
F1	BBRv3	0	4.86	4.84	0.997
F1	BBRv3	5	4.62	5.11	0.991
F1	BBRv3	10	5.83	3.86	0.862†
F1	BBRv3	15	3.51	6.17	0.866†
F1	BBRv3	20	4.62	5.10	0.987
F1	BBRv3	30	4.62	5.06	0.991
F1	CUBIC	0	1.71	7.81	0.713
F1	CUBIC	10	1.86	7.80	0.734
F1	CUBIC	20	1.14	8.32	0.655
F3	CUBIC	0	3.43	6.18	0.888

† High-variance 2-run average; individual runs: lead=10: $J = 0.738$, 0.986; lead=15: $J = 0.742$, 0.991. Polarity reverses between runs, indicating random transient ordering rather than a systematic bias.

A. Limitations and Future Work

Orbit table staleness. The 10-second min_rtt expiry causes BBR-SAT to fail for handover times $T_{\mathrm{HO}} \geq 120\,\mathrm{s}$. Refreshing the orbit table entry during the PREDICTED window — a one-line change in the PREDICTED handler — would fix this at the cost of a brief ProbeBW_DOWN that re-seeds min_rtt.

Loss injection. All experiments are zero-loss. Satellite links exhibit correlated loss (rain fade, co-channel interference) that may interact with BBRv3's loss-recovery path. The interaction between the DRAIN phase and is_in_recovery state — currently cleared on PREDICTED — requires evaluation under realistic loss models.

Multi-path QUIC. Satellite multi-path scenarios (e.g., LEO primary + GEO backup with simultaneous active paths) involve per-path BDP state and cross-path pacing interaction not addressed here.

BBR/CUBIC coexistence. The 64/36 CUBIC advantage in steady-state GEO (F3, $J = 0.888$) and the 82/18 split during F1-CUBIC ($J \approx 0.71$) are inherited from vanilla BBRv3's loss-averse behaviour on large-BDP links. Addressing this would require either ECN-based queue-occupancy signalling or a fairness-aware pacing mode in BBRv3 itself.

VII. Conclusion

We presented BBR-SAT, a minimal extension of BBRv3 that equips QUIC with orbit-aware congestion control for satellite inter-orbit handovers. By integrating an ephemeris-seeded orbit parameter table and a two-phase PREDICTED/CONFIRMED signal protocol, BBR-SAT eliminates the 8×RTT stall inherent in BBRv3's bandwidth filter on downward BDP transitions. In a zero-loss simulator spanning all six pairwise LEO/MEO/GEO transitions, BBR-SAT is the only evaluated mechanism to converge consistently on the critical LEO→GEO transition across the full range of tested lead times (0–20 s), achieving T90 = 2.5 s with a 36× reduction in peak queue occupancy relative to vanilla BBRv3. Fairness analysis confirms that advance knowledge does not translate into a bandwidth advantage: BBR-SAT co-exists equitably with uninformed BBRv3 flows ($J = 0.997$) and preserves the existing BBRv3/CUBIC balance.

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